EXNO1
29/1/26

VERIFICATION OF KIRCHHOFF'S LAWS.

AIM:

To verify Kirchoff's voltage law and Kirchhoff's current law both theoretically and practically for a DC circuit.

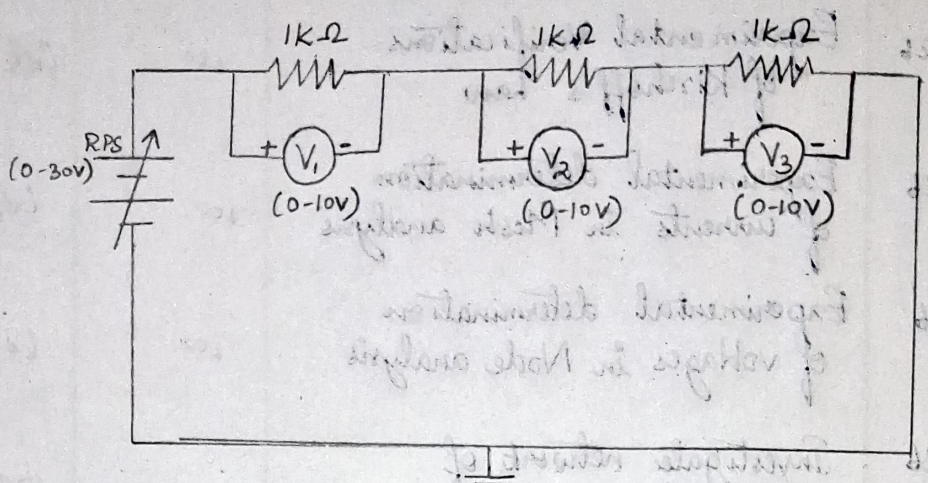
APPARATUS REQUIRED:

S.NO	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power Supply (RPS)	(0-30 V)	1
2.	Voltmeter	(0-30V) MC	3
3.	Ammeter	(0-10mA) MC	3
4.	Resistors	1KΩ	3
5.	Bread board	-	1

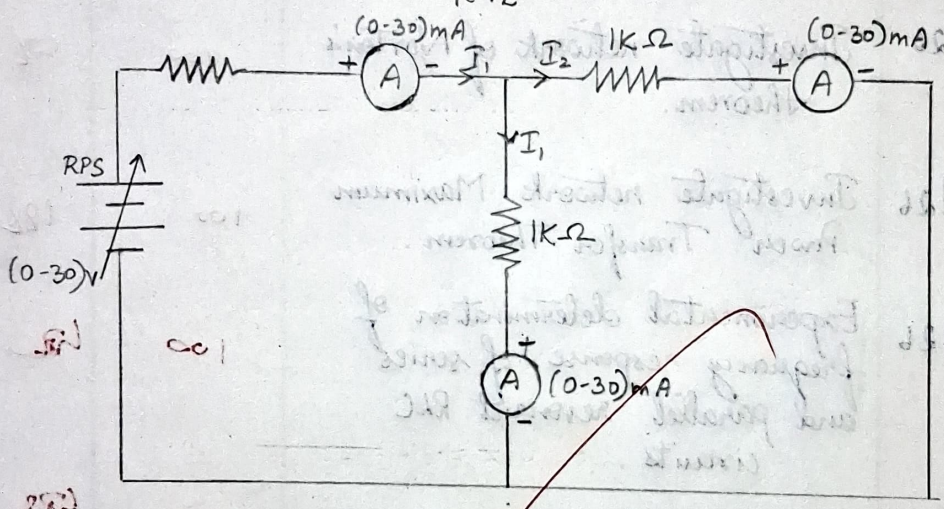
PROCEDURE:

1. Give connections as per the circuit diagram.
2. Switch ON the supply, vary the RPS (Regulated Power Supply) and set a particular input voltage.
3. Note down the readings of ammeters and voltmeters and tabulate them.
4. Vary the RPS for different input voltages and note down the readings of all the meters.
5. Reduce the RPS to its minimum value and switch OFF the supply.
6. Using the tabulated ~~values~~ values, verify Kirchoff's laws practically, and verify it theoretically.

★ CIRCUIT DIAGRAM:



KVL



KCL

TABULAR COLUMN

V (Volts)	V_1 (Volts)	V_2 (Volts)	V_3 (Volts)	$V = V_1 + V_2 + V_3$	I (amps)	I_1 (amps)	I_2 (amps)	$I = I_1 + I_2$ (amps)
2	0.71	0.72	0.71	2.1	4.3mA	2.1mA	2.1mA	4.2mA
6	2.06	2.06	2.06	6.18	12.5mA	6.2mA	6.2mA	12.4mA
9	3.0	3.0	3.0	9.0	18.7mA	9.3mA	9.3mA	18.6mA

CALCULATIONS :

For KVL circuit

i) $V = 2V$

$I = V/R = 2/3 \times 10^3$

$I = 0.71 \times 10^{-3} A$

$V_1 = IR_1 = 0.71 \times 10^{-3} \times 1 \times 10^3$

$= 0.71V.$

Similarly.

$V_2 = IR_2 = 0.72$ $\quad V_1 + V_2 + V_3 = V$

$V_3 = IR_3 = 0.71$ $\quad 0.71 + 0.72 + 0.71 = 2V$

ii) $V = 6V$

$I = V/R = 6/3 \times 10^3 = 2 \times 10^{-3} A.$

$V_1 = IR_1 = 2.06 \times 10^{-3} \times 1 \times 10^3 = 2.06$

$V_2 = IR_2 = 2.06 \times 10^{-3} \times 1 \times 10^3 = 2.06$

$V_3 = IR_3 = 2.06 \times 10^{-3} \times 1 \times 10^3 = 2.06.$

iii) $V = 9V$

$I = V/R = \frac{9}{3 \times 10^3} = 3 \times 10^{-3} A$

$V_1 = IR_1 = 3 \times 10^{-3} \times 1 \times 10^3 = 3.$

$V_2 = IR_2 = 3 \times 10^{-3} \times 1 \times 10^3 = 3$

$V_3 = IR_3 = 3 \times 10^{-3} \times 1 \times 10^3 = 3$

$V = IR$ (Ohm's Law)

$R_T = R_1 + R_2 + R_3 = 1K + 1K + 1K$

$R_3 = 3K\Omega$

Total Current $I_T = I_T = V/R = 2/3 = 0.66mA$

Voltage across each resistors

$V_1 = 0.66 \times 1 = 0.66$

$V = V_1 + V_2 + V_3 = 0.66 + 0.66 + 0.66$

$= 1.9V$ (Theoretical value)

$\boxed{\text{Practical value} = 2V}$

S.NO	Description	Marks alotted	Marks obtained
1.	Experimental Setup / Connections	5 Marks.	20
2.	Observation / Readings	5 marks	20
3.	Calculation / Results	5 Marks	20
4.	Viva	5 marks	20
5.	Record	5 Marks.	20

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RESULT:

Thus Kirchoff's Current Law and Kirchoff's Voltage law are verified practically.